

Specifying the Role of Political Leadership in Energy Policy Change, 1975 – 2006

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**Prepared for ECPR Workshop
“The Determinants of Policy Change: Advancing the Debate”
May 11, 2007
Helsinki**

Introduction: The need to anticipate termination when initiating policy change

This paper explores the role that political leadership can play in energy policy making, focusing on the United States. Energy offers a fertile policy domain from which to revisit Peter Hall's (1993) explanation of how political leadership can enable a policy paradigm shift. His insights into the contribution that political leaders make to replacing policy paradigms were formed by analyzing the 1980s shift in British macroeconomic policy away from Keynesianism and toward monetarism under Margaret Thatcher. A fuller understanding of the dynamics of policy paradigm shift can be gained by contrasting what happens when one political leader tries to move toward new goals and priorities but fails to achieve such a shift, despite significant efforts and is followed by another leader who refocuses the direction of policy change.

Such a false start toward conservation and substitution of imported oil is the story of American energy policy from 1973 through the 1981. After that failure to change the paradigm of American energy use, a new policy of military intervention was launched to keep Middle East oil flowing to the U.S., in ever greater volumes. America's lack of success in finding alternatives to Middle East oil shows, among other things, that any political leader who does not identify policy termination as a necessary concomitant of policy innovation will fall short of a paradigm shift no matter how much effort goes into administrative restructuring and programmatic innovation. It also illustrates that until contending elements of related policies and programs have been terminated, the effort to attain paradigmatic change can easily be rolled back. This paper seeks to explain why political leadership should prepare for policy termination from the outset of any initiative to pursue paradigmatic policy change.

Bardach opened the door to studying policy termination when he suggested that it be viewed as "... a special sub-class of ... policy adoption contests [in which] distinctive coalitions generally form on both sides ... and that termination contests are usually more bitter and harder to win, than most policy adoption contests." (1976: 131) The obstacles to policy termination have been presented as formidable indeed. These include cognitive

barriers such as the modern psychological orientation in which “people do not readily confront issues pertaining to death” (deLeon, 1978: 287) as well as “the general moral repugnance Americans (as well as many other peoples) feel towards the deliberate disruption of arrangements which people have learned to rely on for a significant portion of their livelihoods or careers.” (Bardach, 1976: 128) They also involve the political mobilization created through “the influence that policy has upon politics” in Lowi’s classic formulation. (1972: 301) Any policy will create material and status benefits for constituencies that may or may not have been previously organized, but which can be counted on to mobilize and fight fiercely against termination.

deLeon notes that “anti-termination coalitions” are forces to be reckoned with because:

... the internal bureaucracy that would be opposed to termination actions is well entrenched (which Bardach (1977:128) defines to mean ‘having easy and regular access to powerful governmental allies’... [and] external allies can be particularly purposeful because they will have already benefited from the threatened agency’s services [whereas] groups politicking for the initiation of a program... can only lobby on the basis of expectations. (deLeon, 1978: 291)

Significant authority is thus needed to impose losses on such highly mobilized political defenders of existing policies and programs. Coate and Morris (1999) formalized the economic logic motivating beneficiaries of an existing policy to increase their willingness to pay for policy maintenance efforts (and its benefits) into the future and hence become regular advocates for “policy persistence.” Something beyond charm and persuasion is needed to overcome such resistance to changing policy.

In exploring the dynamics of “policy succession,” Hogwood and Peters (1982) assert that successfully managing a transition from old to new policy requires considerable bureaucratic management skills in order to resolve conflict across organizational boundaries, more so than is the case when initiating policy. They point out that:

... implementing policy succession is more likely to involve greater interactive efforts with other policies and programs than is genuine innovation. Therefore, the interorganizational politics of accommodation and adjustment are more likely to be pronounced in policy succession than in policy innovation. (Hogwood and Peters, 1982)

Political leaders should thus expect opposition to arise from policy participants both inside and outside government and be ready to deal with it. In theory, political leaders should have more leverage over the bureaucratic agents they formally control. But in practice, as devotees of the BBC series “Yes, Minister” will recall, internal opposition to policy succession can be even more entrenched and intransigent than the protest of organized societal interests. This is why leaders ought to consider challenging the intellectual and organizational roots of those bureaucracies that would oppose termination as a key element of any attempt at policy succession. Bureaucracies responsible for policies that relate to the desired area of change must be cleared out of the way to open the path for significant innovation.

Brooker (2005) draws upon Schumpeter’s understanding of economic innovation to emphasize the leader’s role in launching waves of creative destruction inside the state in order to make room for the policy initiatives that combine new factors of production and organizational arrangements to yield policy innovation. Brooker notes that the “pioneering leader” must possess the “... intellectual will power to overcome psychological habits of thought, and the emotional will power to overcome social resistance.” (2005:22) Thus, when the pioneering leader sets a course for policy change, she should also recognize the need to terminate policies that are likely to facilitate organized opposition and reinforce the cultural resistance to change. She must also know where and how to clear away the administrative underbrush that provides cover for the defenders of “business as usual” within government.

When it came to creating a policy framework that could successfully launch a new policy paradigm akin to the monetarist revolution in macroeconomic management, such anticipatory policy termination would have required considerable pruning back of

America's housing and transportation policies that encouraged the urban sprawl and "auto dependence". (Newman and Kenworthy, 1999) While there was some effort to link housing and transportation policies with energy policy during the 1970s by *adding* new programs, such as automotive fuel efficiency regulations, there was little or no effort to remove or change the established policies that generated future demands for additional oil consumption. These included the Interstate Highway program, the Airport and Airway development program, home mortgage subsidies and tax expenditures for new suburban development. The lack of success in policy termination in housing and transportation turned out to be a missed leadership opportunity that virtually guaranteed America's new energy policy would fall short of a paradigm shift.

Energy Policy Change from 1973 to 1981: Policy Innovation Without Termination

The political context framing America's first attempt to develop a new energy paradigm bears some similarity to that surrounding Great Britain's monetarist macroeconomic restructuring. Indeed, the driving forces that brought policy initiatives to the threshold of paradigm shift in these two areas had common roots. An unanticipated change in energy availability had arisen that challenged both the efficacy of existing living arrangements, and the economic structures that were expected to create wealth in western societies. During the 1970s, the affordability and availability of oil became a policy problem of the first order. Over the next ten years, energy would join macroeconomic management as a litmus test for politicians aspiring to any position of leadership.

It is worth considering what energy policy alternatives would have looked like from the perspective of those facing America's first energy crisis during the 1970s. There were at least three major paradigm shifts in contention at that time. Orr (1979: 1030) characterized the debates over energy policy during the 1970s as "the first attempt to manage politically the transition from one energy base to another." One policy paradigm that was advanced in response to the energy crisis identified a clear and narrow goal of *increasing energy supply*. Energy policy supply-siders came to their faith in increasing energy inputs by rejecting all alternatives. The focus on supply implied that America's

high energy intensity was both unavoidable, and desirable. Restraining energy consumption through taxation or regulation was seen to be politically illegitimate, economically dangerous and technologically self-defeating.

An apparent paradox in the logic behind the supply side paradigm was the purported fragility of society in the face of any reduction of existing energy inputs, coupled with an anticipated resilience to the future risks that would accompany ambitious efforts to boost energy supply:

On the one hand technologically advanced societies are portrayed as so inflexible that any leveling or reduction in energy consumption would cause disaster. On the other hand, these same societies are portrayed as flexible enough to withstand the acknowledged certainty of periodic catastrophes, including large oil spills, nuclear meltdowns, liquid natural gas explosions and global climate change (Orr, 1979: 1031)

Such logic was embodied in the core beliefs of a major “advocacy coalition” as defined by Sabatier and Jenkins Smith (1993) comprised of energy producers, automotive and aerospace manufacturers, home builders and real estate developers, among others. The disparate combination of fragility and resilience noted by Orr was not inconsistent from their perspective, since the costs of disruption in oil supplies would be concentrated on these interests, while the risks of an aggressive increase in future energy supplies would be diffused much more widely across society, given the uncertainty of exactly where and when a future energy-related accident might claim its victims.

An alternative paradigm that gained credence in the 1970s identified *decoupling* energy use from economic development as the key to solving the energy crisis. Here, government efforts would seek to reshape the consumption of energy by creating incentives to improve efficiency and diversify sources. Government’s role would be expanded in this paradigm, in a way that the supply side paradigm option rejected. While the supply-side paradigm would support greater authority and additional public resources being used to support the efforts of corporations and the military-industrial

technostructure, both to seek additional oil and experiment with new energy sources (e.g., breeder reactors, cold fusion), the decoupling paradigm would require deepening government's relationship with citizens in matters of energy. From regulating energy efficiency to building public awareness of the benefits from conservation, the decoupling paradigm would imply both a new kind of public sector activism and a new kind of citizen engagement.

While both the supply side and decoupling paradigms endure as recognizable as energy policy alternatives that are debated to this day, the *energetics* paradigm option is less familiar. This “soft” approach to reorienting the use of energy posited that the relationship between energy and human behaviour was both interdependent and amenable to policy innovation. New energy alternatives are thus seen to require new political arrangements in order to yield a more effective outcome than either the goals of increasing supply or reducing demand that would be pursued through existing governance arrangements. Energetics conceives of energy policy “... as a lever to move society toward a more sustainable, if less extravagant, basis while enhancing equity and participation.” (Orr, 1979: 1036)

An illustration of how the energetics paradigm would differ from the decoupling paradigm can be found in their approach to introducing new energy sources. In decoupling, new energy sources were seen as one of several means for rebalancing the relationship between energy consumption and economic growth. But from an energetics perspective, shifting to diffuse energy production by substituting solar, wind, and geothermal sources for fossil fuels offer an essential political contribution to a new paradigm. Such alternative sources could be controlled by local governments, cooperatives, or community based organizations rather than the large energy corporations that dominate the production of fossil fuels. Smaller scale energy production would reshape politics by empowering small groups of citizens to make diverse and ultimately more resilient, decisions on their levels and sources of energy need. Supply-side, decoupling, and energetics paradigms influenced debate over how far, and in what direction, U.S. policy innovations should go. And while there were numerous options

vying for a place on the energy policy agenda, there was no doubt that a political leader was in need of an answer to the question of which energy policy was right for America in order to be taken seriously.

Just as elected officials could not remain credible without an economic strategy to address rising inflation and unemployment levels, they were also now expected to have a plan for ensuring affordable energy to enable economic growth. Kemezis and Wilson have labeled the ensuing series of policy initiatives undertaken by the U.S., Europe, and Japan as an “energy policy explosion” pointing out that:

By the end of the 1970s energy was treated as an essential, almost precious commodity, necessary to a nation’s well being and thus the stuff of high politics and national security. In the absence of any serious or sustained effort at formal international energy decision making to replace the formal function of the oil companies, virtually every importing country in the world reacted to the new conditions and concentration of power in OPEC in a strikingly similar way, pushing deeper and deeper into national and international energy policymaking on its own. (Kemezis and Wilson, 1984: 3 - 4)

From the outset of this energy policy explosion, two alternative approaches were discernable in the policy leadership efforts of Presidents Nixon, Ford, and Carter. Nixon sought policy that would offer short term relief from the economic pain produced by the oil shock. Policy was initiated to restrain rising oil costs and stimulate domestic production, basically seeking to guide America’s energy supply back toward the *status quo ante* that had existed before the Arab oil embargo. Presidents Ford and Carter introduced progressively more ambitious policy options, seeking to restrain the demand for oil through conservation and development of alternative energy sources.

The short term policy focus under Nixon precluded the need to consider termination. The Nixon administration’s incapacity for developing a major energy policy initiative that would have raised the need for termination was just as well, since the beleaguered President had no political capital left to devote to such efforts. Jones (1979: 166) writes

that “It is not unreasonable to conclude that the erosion of Presidential authority seriously delayed the organizational and programmatic adjustments to the new energy imperatives.” Gerald Ford attempted leadership that could have led to policy succession and did introduce two new instruments of second order energy policy change. But the Ford administration did not pursue the policy termination that would have been needed to create a clear path for an energy paradigm shift. And President Carter staked a lot of his credibility in domestic policy on comprehensive policy transformation, but failed to overcome entrenched resistance from the energy producers and transportation manufacturers. The search for energy policy solutions began when America’s supply of oil dropped below the level that could meet current demand, triggering what would become known as the “energy crisis.” While this period of policy initiative is generally considered to have begun with the Arab Oil embargo of 1973, the reality was more complicated.

While the Arab oil embargo began on October 19, 1973, when the Organization of Arab Petroleum Exporting Countries (the Middle Eastern subset of the OPEC members) cut off exports to the United States, the Netherlands, and other nations supporting Israel in the 1973 Arab – Israeli war, America was *already* running short on oil supplies. Taylor and Van Doren (2003) note that “By May of 1973 (five months before the embargo), 1,000 service stations had shut down for lack of fuel and many others had substantially curtailed operations.” The legislation put forward to resolve this oil shortage, which would become known as the Emergency Petroleum Allocation Act of 1973, was introduced in the Senate on April 13, 1973, although it was only signed into law following the Arab embargo. Something more than an embargo was behind this element of energy policy innovation.

A strong case can be made that the “problem” which triggered policy innovation efforts was peak oil, a geological term for the point at which the output from a given oil reservoir reaches its maximum. This problem was compounded by the policy intervention of the 1973 Emergency Petroleum Allocation Act, which established a regulatory scheme that presumed domestic oil production could be increased by creating higher pricing for

“new” supplies. If no more oil was available, then higher prices would do little to stimulate its production.

U.S. oil production in the lower 48 states had reached a peak in 1970, although this was not widely recognized at the time. (Ivanhoe, 1996) The peak in U.S. production illustrates an important, yet often contested, geological principle of oil extraction that was first elaborated by M. King Hubbert. Hubbert (1949) hypothesized that once half the oil in any reservoir is extracted, the rate at which the remaining oil can be produced begins to decline. He grounded his explanation of what became known as the “Hubbert curve” on the physics and chemistry of petroleum formation, extraction and refining. Hubbert then extrapolated his theory from the logic governing a particular oil field to one that would govern the extraction and eventual depletion of total petroleum reserves, predicting that global oil production would inevitably peak and then decline asymptotically to near zero. The policy implications of a global peak in oil are discussed below, but in the early 1970s America first confronted the implications of its inability to meet the demand for petroleum from domestic sources.

The Emergency Petroleum Allocation Act of 1973 sought to achieve two energy goals – low prices and expanding supply, which proved incompatible after the total production of American petroleum in the lower 48 states had begun to decline. The legislation continued a price ceiling on “old” oil coming from wells already in production. This cap had been introduced as part of wage and price controls in 1971. While the price cap was initially intended to moderate the inflationary effects of energy pricing, it took on greater political momentum following the Arab embargo, when the rapidly rising price of oil on international markets would have generated a huge windfall profit for big oil companies in the event that prices were suddenly deregulated. The legislation did exempt “new” oil that was generated by wells put into production after 1973 from any price regulation. This incentive was expected to stimulate domestic oil production by creating the incentive to explore and extract new sources.

Because of Hubbert's peak, the economic incentive to produce new domestic oil created by the Emergency Petroleum Allocation Act of 1973 proved to be no match for the geological reality that there wasn't any unexploited reserve of conventional oil within the lower 48 states that could be quickly tapped to boost America's energy supply. The first policy intervention to address the energy crisis offered little more than short term price relief. Its paradigmatic orientation toward increasing domestic supply was at odds with the geological reality:

As price controls on United States-produced oil continued, the lower prices brought the logical result: the United States increased the quantity of its oil imports. The low prices shielded consumers somewhat from the rise in international prices, and thus did little to encourage conservation; they did even less to encourage domestic production. (Basile, 1982)

The Arab oil embargo thus followed, rather than triggered, a shortfall in America's oil production relative to demand. The embargo had its greatest impact on oil prices outside the U.S., which more than tripled between 1973 and 1974.

While the Nixon administration was being drawn farther away from any normal mode of policymaking by the Watergate scandal, municipal government leaders took some of the first steps to deal with increasingly disruptive energy shortages, providing the closest policy expression of an energetics paradigm approach during the 1970s. Rudel (1980) notes that by January of 1974, New York City police officers had been assigned to keep the peace at gas stations where long lines, limited supplies, and drivers' desperation had led to regular incidents of violence at the pumps. On February 4, 1974, the Mayor of Elizabeth, New Jersey, became the first American elected official to mandate gasoline rationing since the Second World War (Rudel, 1980: 199) Compared to the policy paralysis in Washington during Nixon's final six months in office, local leadership had delivered immediate results that at least managed to deal with the scarcity of gasoline.

As order returned to local gas stations, and odd-even day gas rationing kept energy availability within reach of demand, the role of peaking domestic oil production was

obscured by much more visible diplomatic maneuvering to address the Arab oil embargo. A very modest step toward the energetics paradigm had thus been shown to be possible at the onset of America's energy crisis. Local democracy came up with a formula for allocating available gasoline in ways that national leaders could not address at the time.

By the time that Gerald R. Ford became President, pressure for national energy solutions was considerable. Both energy shortages and the infamous "stagflation" that combined high unemployment with inflation were seen to require a break with past policy. In his first State of the Union Address, President Ford stated that "A massive program must be initiated to increase energy supply to cut demand, and provide new standby emergency programs to achieve the independence we want by 1985." (Ford, 1975) The President's willingness to act was underscored by his announcement of an immediate \$1 barrel increase in the duty on imported oil, rising to \$3 within two months. This amounted to a 30% increase in the price of imported oil. According to energy economist Philip Verleger Jr., "Gerald Ford was the first U.S. president to really use the levers of the presidency to try and break our addiction to oil." (quoted in Friedman, 2007)

The Ford administration introduced two new policy instruments which continue to influence American energy policy to this day. In Hall's terminology, these second order policy changes were included in the Energy Policy and Conservation Act of 1975 (EPCA). This legislation introduced Corporate Average Fuel Economy regulations to mandate automobile fuel efficiency improvements that would take the fleetwide consumption from 18 miles per gallon in 1978 to 27.5 miles per gallon in 1985. EPCA also created the Strategic Petroleum Reserve (SPR), and authorized the storage of up to 1 billion barrels of crude oil for civilian needs and 300 million barrels for military needs to be built up as a hedge against future supply interruptions. The new policies sought to build new supply side capacity through the SPR and move the automotive sector toward decoupling the growth in vehicle use from future energy consumption.

Creating a regulatory regime for auto fuel efficiency and a strategic petroleum reserve were major policy initiatives. But it is important to recognize what EPCA did not

attempt, along with its accomplishments, in order to highlight the avoidance of policy termination that occurred during this attempt to change policy. ECPA avoided the policy options that would have required changing established programs and organizations. These included: ending the regulation of oil prices established in 1971; raising the federal excise tax on gasoline, which had been set at the time the Interstate Highway program was launched in 1956; changing the federal excise tax rate on new automobiles to introduce a stiff increase on “gas guzzlers.”

Both fuel economy regulation and the strategic petroleum reserve entered previously unoccupied policy space, in that no other legislation or administrative structures existed to address these issues. The opening of a policy window for introducing automotive fuel efficiency regulations, and the subsequent closure of that window during the 1980s and 1990s, are detailed by Perl and Dunn (2007). The result of such an approach to policy change was to create initial opportunities for innovation, but to also limit subsequent efforts to pursue a paradigmatic change regarding energy use and sourcing. Without terminating existing policies such as: oil price controls, highway and airport expansion programs, and; housing subsidies that encouraged sprawl and auto-dependence, there was only so far that energy policy could advance toward a new paradigm based around either conservation or a new relationship between energy and society. Ford’s successor would have both the motivation and the opportunity to seek a more paradigmatic shift in policy.

Jimmy Carter should have been positioned to take energy policymaking to the third order of change and establish a policy paradigm that would have been analogous to the macroeconomic paradigm shift launched by his British contemporary, Prime Minister Thatcher. Carter’s Presidential mandate was built upon an electoral repudiation of the party and ideology that had preceded him. Richard Nixon’s disgrace had undermined more than Gerald Ford’s claim on the Presidency. Both the House of Representatives and the Senate had seen widespread defeat of Republican incumbents. The November 1974 Congressional elections returned unusually large “post-Watergate” Democratic majorities, particularly in the House of Representatives. Democratic representation in the lower chamber had jumped from a lead of 239 to 192 before Watergate to a margin of

291 to 144. Washington appeared poised for a political realignment that would enable paradigmatic policy change. Yet what happened instead was a policy paralysis fueled by a power struggle between Congress and the White House that extended the rivalry between branches of government that had been made acute by Watergate.

Carter came into office determined to change paradigms on several fronts. From his Georgia political career, he brought to the White House "... a conception of political leadership ... in which the virtuous leader articulated the interests of the vast, unorganized public against the demands of groups" (Hargrove, 1988: 4) Energy was a policy domain where such an approach could yield major change. Less than two weeks following his inauguration, Carter signaled his intention to introduce a sweeping energy program, stating:

One of our most urgent projects is to develop a national energy policy.... The United States is the only major industrial country without a comprehensive, long-range energy policy. (Carter, 1977a)

This search for America's new energy policy involved more than new goals relating to energy. Jimmy Carter had been elected on a platform that promised Americans a new mode of national governance, "... one that would dispense with the tired bickering of the old parties, transcend special interest pressures, and generally authorize smart people to generate public policies in the national interest." (Skowronek, 1993: 364) Carter's aim of replacing traditional partisan and bureaucratic politics with "outside the Beltway" efficiency and honesty was typical of what Skowronek identifies as a late stage Presidential leadership regime – one where the challenges of governing appear sufficiently formidable that only a break with the old order could yield results.

Short of fully repudiating that old order, which in Carter's case would have amounted to breaking with the Great Society as his successor Ronald Reagan would eventually do, late regime Presidencies tended to appeal to apolitical values such as competence and honesty to deal with an increasingly dysfunctional administrative environment. According to Skowronek (1993: 365), "This leadership stance recurs in presidential

history because outmoded regimes have an institutional staying power and because the technocratic appeal has a momentary resonance in one that is lingering on in disarray.” Such a technocratic orientation to policy development would be quite open to third order change, and presumably comfortable with terminating previous policies that were seen as dated, ineffective, or otherwise suboptimal.

Carter launched a policy development process that aimed for non-incremental change by directing his expert energy advisor (and future Cabinet secretary) James Schlesinger to develop a sweeping new policy that could be unveiled to Congress and the American people within 90 days. Although Schlesinger asked for more time to consult broadly, Carter was determined to avoid the delay and compromise that would arise from interest group politics. The policy package under development would be presented to Congress as a matter of necessity, rather than negotiated with advocacy coalitions and the Congressional leadership because Carter believed that “compromise was acceptable only after an all-out effort to sell the optimal policy had failed.” (Hargrove, 1988: 13)

Jimmy Carter was counting on his Presidential “honeymoon” and public concern about energy to provide a window of opportunity to enact his energy program. A policy breakthrough analogous to Franklin Roosevelt’s launch of the New Deal during his “first hundred days” was the implicit model of policy change. But such an exercise of Presidential authority had only been possible after the economy had collapsed and corporate America became widely regarded as incapable of restarting America’s engine of growth.

Jones (1989) has posited that a key variable in effective leadership is the division of labour between political elites and economic elites. Political elites must account to the public through periodic elections that render judgment on how successful they have been in creating opportunities for advancement. In America, these opportunities are expected to be perpetually improving, as Ronald Reagan reminded the electorate when he bested Carter in the 1980 Presidential debate by asking citizens “Are you better off now than you were four years ago?” Economic elites are essential for delivering the economic

growth that enables a politician to claim credit for economic opportunities and thus expect to "... get special attention in the councils of government." (Jones, 1989:6) Thus the type of leadership that is possible will depend "... basically on the strength of political elites relative to economic elites." (ibid.)

In 1977, although times were tough and energy was an acknowledged contributor to economic stagflation, America as a whole was not seen to be at risk of economic failure. What emerged from the Carter administration's first 90 days was an energy program that would have required significant policy termination to yield a bold departure from previous policy that could well have led to paradigm shift. The proposal for policy change was bold, but the details were so complex that "no one besides Schlesinger could explain the total package." (Davis, 1979: 192) yet, the U.S. Congress was expected to endorse the entire package as a "silent partner" in the President's policy transformation. (Jones, 1979: 172) With vigorous opposition from the energy and transport sectors, among others, this was a recipe for failure.

President Carter launched his plan by seeking to rally the American people in a nationally televised address on April 18, 1977 that became famous for its declaration that transforming the ways America used energy was "the moral equivalent of war". The President began bluntly:

We must not be selfish or timid if we hope to have a decent world for our children and grandchildren. We simply must balance our demand for energy with our rapidly shrinking resources. By acting now, we can control our future instead of letting the future control us. (Carter, 1977b)

Carter then called upon the American people to recognize their energy inefficiency, and do something about it:

Ours is the most wasteful nation on earth. We waste more energy than we import. With about the same standard of living we use twice as much energy per person as do other countries like Germany, Japan, and Sweden.

Our decision about energy will test the character of the American people and the ability of the President and the Congress to govern. This difficult effort will be the ‘moral equivalent of war’ – except that we will be uniting our efforts to build and not destroy. (Carter, 1977b)

The Carter energy plan would have required a paradigm shift in decoupling from America’s wasteful past to a future where energy efficiency would buy the time needed to introduce alternatives to oil. These new sources were left somewhat open, but ranged from familiar sources that had been improved upon such as “clean coal” to radically unfamiliar sources like solar power. The Carter plan implied a shift away from corporate influence and direction over energy and toward a new energy partnership directly between government and citizens. This was exemplified by the program’s first principle, “... we can have effective and comprehensive energy policy only if the government takes responsibility for it and if the people understand the seriousness of the challenge and are willing to make sacrifices.” (Carter 1977b) Shared sacrifice was the core value behind Carter’s proposed transformation of the American energy paradigm.

After months of political skirmishing in the legislative trenches, Carter lost control over the direction of America’s energy policy and the principal sacrifice that was made was that of Carter’s conservation goal. Policy termination was not a part of the legislation, leading Ponder (1999: 106) to conclude that while President had initiated new energy policy “it had been stripped of many of the substantively important points that Carter had wanted to retain.” Not surprisingly, the results fell short of the third order change in policy goals. The Department of Energy (DOE) was created, but focused adjusting the existing policy instruments and programs that were brought together under its administration.

The DOE’s incremental approach to policy implementation did little to prepare the U.S. for its second oil shock in 1979, when the Iranian revolution deposed America’s closest ally among Middle East oil producers and the global supply of oil was curtailed, America was again beset with gasoline shortages and price spikes. Jimmy Carter sought to launch

an energy policy transformation for the second time, which yielded another signature phrase of his presidency stated during a national address on July 15, 1979. In that speech, Carter claimed the reason why America had been unable to resolve its energy crisis was because of a deeper “crisis of confidence” which:

... strikes at the very heart and soul and spirit of our national will. We can see this crisis in the growing doubt about the meaning of our own lives and in the loss of a unity of purpose for our nation. The erosion of confidence in the future is threatening to destroy the social and political fabric of America. (Carter, 1979)

The President blamed government for the policy paralysis that had allowed America’s problems to deepen, pointing out:

What you see too often in Washington and elsewhere around the country is a system of government that seems incapable of action. You see a Congress twisted and pulled in every direction by hundreds of well-financed and powerful special interests. You see every extreme position defended to the last vote, almost to the last breadth by one unyielding group or another. You often see a balanced and fair approach that demands sacrifice, a little sacrifice from everyone, abandoned like an orphan, without support and without friends. (Carter, 1979)

Carter’s lament for the public interest suggested some need for an energetics paradigm, in which a change to America’s mode of governance would enable the transformation of future energy usage. While the form that such a political renewal would take remained unstated, the proposed energy policy changes that were proposed specifically embraced the decoupling paradigm:

I am tonight setting a clear goal for the energy policy of the United States. Beginning this moment, this nation will never use more foreign oil than we did in 1977 – never. From now on, every new addition to our demand for energy will be met from our own production and our own conservation. (Carter, 1979)

An energy security corporation was proposed to produce energy that would replace 2.5 million barrels of imported oil by 1990. This new instrument became the Synthetic Fuels Corporation. An energy mobilization board was also proposed. This entity was meant to be a catalyst for policy change, “like the War Production Board in World War II [it] will have the responsibility and authority to cut through the red tape, the delays, and the endless roadblocks to completing key energy projects.” (Carter, 1979)

The President obtained more of his preferred policy options from this second round of energy policy initiation, although Congress did not authorize the energy mobilization board. Some saw this improvement over the first round of energy policy innovation as an example of “Presidential learning.” (Ponder, 1999) But the reach of the policy leadership was limited by Carter’s defeat at the polls by Ronald Reagan. Electoral termination had cut short the potential for policy termination, with one energy policy scholar concluding that:

Carter ... may have overestimated the willingness of the people of the United States to engage in self-sacrifice. The conventional wisdom is that Carter’s description of the United States as ‘the most wasteful nation on earth’ in terms of energy use contributed to Ronald Reagan’s election. (Romm, 1991)

The Reagan administration wasted little time in reversing most aspects of the Carter administration’s approach to energy policy. Decoupling and energetics were both anathema to Reagan’s laissez-faire ideology which sought limit government’s role in the economy and society in areas where corporate America could deliver alternatives. Energy was certainly one such domain.

Energy Policy from 1981 – 2000: Maintaining an Old Paradigm Through Domestic Termination While Launching a New One Through International Military Intervention

Although the term “Reagan revolution” became the most common label for a range of policy transformation during the 1980s, neither the President himself nor some of the leadership scholars who analyzed his influence on American politics saw the dynamic of

change as revolutionary. Skowronek (1993: 409) quotes Reagan as reflecting “They called it the Reagan Revolution and I’ll accept that, but for me it always seemed more like the Great Rediscovery; a rediscovery of our values and our common sense.” The Reagan presidency was much more of a counter-revolution than anything else, an effort to dismantle America’s capacity for the kind of big, active government that would have been needed to pursue an energy decoupling paradigm envisioned by Carter.

Facing the ills of modern governance as an outsider, Carter failed in his crusade to redefine the political order in Washington. Ronald Reagan found enduring popularity by working within that order to decimate government instead. The premise of the Reagan “revolution” was that repudiating most domestic public sector activities and accomplishments that had been launched since Dwight Eisenhower (and possibly Herbert Hoover) would enable civil society and free enterprise to recover a rightful place in leading America back to a path of perpetually expanding opportunities that had been its due. In Reagan’s America:

Nightmarish projections of the future were to be dispelled; Carter’s ‘no easy answers’ the journalists’ predictions of national decline, the economists’ prophecy of a new age of limits and individual sacrifice – all were forthrightly rejected. By releasing the creative energies of the private sector from their government shackles, Reagan promised to reset the clock, to make it once again ‘morning in America.’ (Skowronek, 1993: 415)

When it came to using energy at home, the Reagan administration wasted little time in stepping back from the pursuit of conservation or alternatives to oil. Termination of much policy that had been created since 1973 came quickly. Policy instruments from Nixon’s oil price cap to Ford’s automotive fuel efficiency regulations to Carter’s oil import quotas were either rolled back or abolished outright. The fate of the Synthetic Fuels Corporation is illustrative of the Reagan administration’s approach to policy termination. When the corporation was abolished in 1985, Crawford (1986: 18) characterized it as “Wracked by charges of mismanagement and paralyzed for a period by the lack of an operating quorum

on the board of directors,” noting that “the corporation committed just \$1.3 billion of its original \$14.9 billion in spending authority.”

By the close of Reagan’s first term, America’s policy initiatives to decouple energy use from economic activity had been largely abandoned in favour of supply side initiatives. Rather than seeking energy solutions within the U.S. that involved any shift away from oil, energy policy would focus beyond America’s borders to tap oil reserves that were far from their production peak. The decline of oil prices during the late 1980s and early 1990s, spurred by a massive new production from the North Slope of Alaska, the North Sea and Mexico made it easy to reject Hubbert’s earlier theory of peak oil as heresy. There was thus no need to maintain any aspect of the Carter energy policy initiatives, with one important exception.

The one inherited energy policy innovation that the Reagan administration eagerly embraced was the Carter Doctrine. Responding to the Soviet invasion of Afghanistan in December of 1979, Carter had declared that Persian Gulf oil access was now a part of America’s sphere of influence:

In January 1980, the Carter doctrine identified the Persian Gulf as a ‘vital interest’ of the country and declared that ‘an attempt by an outside force to gain control of the Persian Gulf region would be regarded as an assault on the vital interests of the United States.’ The Carter administration subsequently established a Rapid Development Force for use in the Middle East, deployed a permanent naval force in the Persian Gulf, and acquired new military bases in the region. (O’Rourke and Connolly, 2003)

This doctrine was incorporated into the Reagan administration’s defense buildup. Secretary of Defense Caspar Weinberger issued a classified Defense Guidance report that stated “Whatever the circumstances, we should be prepared to introduce American forces directly into the region should it appear that the security of access to Persian Gulf oil is threatened.” (Stork & Wenger, 1991) Based on such logic, the Rapid Development Force was expanded into the “U.S. Central Command” (Centcom) putting its organization on a par with the largest joint service command structures of the United States armed forces.

Centcom has a geographical focus on the Middle East and North Africa, but was strongly influenced by the focus on energy that it inherited from the Rapid Deployment Force. Klare (2004: 18) asserts that “Centcom’s principal mission is to protect the flow of oil from the Persian Gulf to the United States and American allies around the world.” Although the U.S. initially “... came up against the reluctance of Arab rulers to link themselves openly with the chief Western power and the main ally of Israel” agreements were quietly made with Egypt, Oman, and Bahrain to provide air and naval bases that could enable effective policing of the sea routes through which so much of the world’s oil passed. (Stork and Wenger, 1991: 23)

Installing America’s petroleum armada in the Persian Gulf was not cheap. Romm (1991: 31) estimated that “In 1985 the United States paid more than \$200 per barrel of Gulf oil just to keep the shipping lanes open – seven times what it paid for the oil itself.” The cost effectiveness of such a policy initiative designed to maintain the uninterrupted flow of Middle East oil, could not be compared to energy conservation or decoupling options within the U.S., since these initiatives were being wound down. When the Reagan administration submitted a required “National Energy Plan” to Congress in 1986, it emphasized the success of “energy policy planning” that had enabled producers to develop new supplies compared to the “micromanaged energy planning” which had sought to have government lead in influencing demand. (Oil and Gas Journal, 1986: 44) Nothing was said about the growing intervention in the Persian Gulf that was becoming the most active element of U.S. energy policy.

Between 1985 and 1989, America’s oil imports from the Persian Gulf grew by over 500%, from under 100 million barrels to over 600 million barrels per year. (Romm, 1991) Middle east oil imports and military activity have tracked one another through the Reagan and Bush administrations. Anxiety regarding the protracted war between Iran and Iraq was seen to have persuaded the Saudi monarchy change its position against basing U.S. military in the kingdom. During the 1980s, \$14 billion was spent on building bases for American ground troops and a \$50 billion Gulf-wide air defense system was installed

in Saudi Arabia (Lefebvre, 2003). These military facilities were put to full use during the first Gulf War in 1990 – 1991 and remained operational during the Clinton administration, making it easy to scale their activity up again during once the second Gulf War began in 2003.

Writing just prior to the September 11 terrorist attacks, Klare pointed out that this new energy security paradigm generated its own risks:

... while American might may well have discouraged some potential challengers from playing a more assertive role in the Gulf area, the very magnitude of American military presence – along with Washington's periodic use of force – has also provoked opposition to U.S. policies. (Klare, 2001: 53 – 54)

The international military initiatives that were carried out during America's second wave of energy policy change during the 1980s and 1990s have been far more ambitious and costly than any of the efforts to reshape domestic energy use during the 1970s were.

Concluding Thoughts on Termination and Change in U.S. Energy Policy

Unlike the British paradigm shift in macroeconomic management analyzed by Peter Hall, America's attempts to reformulate energy policy have been less cohesive, with a considerable difference emerging in the nature of change between domestic and international policy. During the 1970s, responses to America's energy crisis first moved toward a paradigmatic shift in domestic policy. Policy innovation proved to be most effective at the local scale, where municipal and state government leaders instituted temporary gasoline rationing schemes to manage scarcity. New goals of upholding public order and equitable access to a limited resource were introduced into what had previously been a private commercial transaction.

Policy change at the federal level began by adjusting existing instruments, such as oil import quotas and the regulated price levels for oil, but proceeded to adding new

instruments such as the CAFE regulatory regime to improve automotive fuel efficiency and the Strategic Petroleum Reserve to cushion future fluctuations in oil supply. While these initiatives created new policy niches and introduced measures that could facilitate a transformation in the goals for America's energy use, they did not yield a paradigm shift. Efforts to decouple economic growth from oil consumption and to introduce alternative energy sources were unsuccessful because they remained at odds with established policies in transportation and housing that continued to demand more oil. The advocacy coalitions supported by these established policies proved successful in resisting alternative energy policy goals, portraying them as emphasizing the un-American value of self-sacrifice.

The effort to transform the paradigm for energy use inside the U.S. was repudiated after 1981. Government's role in energy research, regulation, and development of alternative technologies was scaled back considerably under the Reagan repudiation of a public sector activism. At the same time, the deployment of America's armed forces to assure uninterrupted supplies of oil from the Middle East began, initiating a paradigm shift in military intervention to secure oil access. Klare (2004: 18) distils the essence of this paradigm shift by noting that "Slowly but surely, the U.S. military is being converted into a global oil-protection service."

Among the explanations for why the U.S. turned away from a domestic energy paradigm shift and advanced toward a military paradigm where intervention to secure oil imports has become an overarching American security goal, two possibilities stand out as particularly important. First, the leadership that a President can exercise is much greater in the international arena, especially in the role of Commander-in-Chief of the armed forces, than it is in any area of domestic policy. President Carter thus found it easier to successfully introduce the Carter Doctrine in the final year of his administration than he did to establish a comprehensive energy policy over his entire four years in office. His successors, both Republican and Democrat, have found it similarly more attractive to build upon this foreign and military policy initiative than to revisit the pitched policy

battles accompanying any effort to decouple energy use from economic growth or to introduce alternative sources to oil.

The second explanation for why policy change efforts shifted from a domestic to an international focus after the 1970s is more geological than political, and draws upon concept of peak oil. When it became clear that oil production had peaked in the lower 48 states, the only policy paradigm shift that would maintain the economic and political expectations of the dominant advocacy coalition was an embrace of the supply side, and the only place to obtain the bulk of that additional supply was beyond America's borders. Failing to secure the flow of imported oil from the Middle East would thus have added pressure to pursue the decoupling and energetics paradigms, perhaps by delegitimizing the privileged position of corporate elites in energy and transportation. This would have entailed an even higher level of economic disruption than what occurred in the 1970s.

America's political leadership continues a commitment to the supply side energy paradigm by maintaining military initiatives in the Middle East and even expanding them into other oil producing areas. But this paradigm's effectiveness is only as good as the ability of global oil production to keep up with growing demand. By 2006, among the 64 nations that have extracted significant volumes of crude oil, production volumes had peaked in 55 countries. (ASPO, 2006: 11) Thus, just as the peaking of domestic oil production precipitated a period of tumultuous policy initiation, and a conflict over terminating policies that were premised on ongoing reliance upon oil as America's primary energy source, the global peaking of oil can be expected to trigger an even more intense debate over new policy options and alternative energy paradigms. Except in this future round of energy policy change, further expanding oil imports from new locations will not be a realistic option. At that point, which may be closer than many political leaders are prepared to admit, the choice will once again be among paradigms that raise the challenge of domestic policy termination and those policies that rely on military power to secure American access to declining global oil supply.

In 1979, President Carter presented the choice over how America's future energy needs could be met as a "turning point" that offered two paths:

One is a path ... that leads to fragmentation and self-interest. Down that path lies a mistaken idea of freedom, the right to grasp for ourselves some advantage over others.... It is a certain route to failure. [The other path is] the path of common purpose and the restoration of American values. That path leads to true freedom for our nation and ourselves. We can take the first steps down that path as we begin to solve our energy problem. (Carter 1979)

The global oil peak will eventually provide another opportunity for American leadership to guide society on the path that Carter had in mind almost three decades ago. There will no longer be the apparently easy alternative of using America's military to guard access to expanding oil production. In this future policy context, the commander-in-chief will be less tempted to avoid the policy termination required to change America's use of energy.

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